

From Code to Careers: Linking Open-Source Software Signals to AI Skill Demand in the Labour Market

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Abstract

The abstract stands alone as a very short version of the dissertation.

The abstract should state the scope and principal objectives of the project, describe the methods, summarize the results and state the principal conclusions.
(Max. 500 words.)

Declaration of originality

I confirm that:

- This submission is my own work, except where clearly indicated.
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Name: Dafydd James Thomas

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Consent to share this work

- By including my name below, I hereby agree to this thesis being made available to other students and academic staff of the Department of Computer Science, Aberystwyth University.

Name: Dafydd James Thomas

Date: 05/09/2025

Acknowledgment

To whoever has the patience to read this :-)

This section is customary, but not obligatory. It makes a brief statement of thanks to those who have helped.

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Chapter 1

Introduction

[1] In recent times, we are seeing that skills and job requirements are evolving rapidly. As the culmination of various effects, including the development of new technologies and evolving consumer preferences, the labour force must adapt in order to satisfy the new requirements. Given the historic pace of labour demand shifts, traditional methods of forecasting these changes are not well-equipped to handle this blistering, and accelerating, rate of change.

By these so-called "Traditional" methods, I mean that which accompany government or firm-level analysis of the skills market. These depend upon data which are limited in the following ways:

1. They use structured data
2. They collect the data periodically (e.g. Annually / Quarterly)

An important feature of these methods is their approach to modelling the dynamics influencing labour demand. Of key importance is the impact of innovation, and how new technology is estimated to affect skill requirements.

This paper aims to investigate whether Open Source Software (OSS) innovation Signals, derived from unstructured text data, can meaningfully contribute to the forecast of AI-related skill requirements present in job postings.

To break this down, there are a few key ideas being explored:

1. Can open source software data represent information about Innovation in the AI sector?
2. Can AI-related skill requirements be modelled the textual data within job postings?
3. Can we integrate the two data channels to produce a quantified measure of their relationships?

Put simply, if we can measure how similar an OSS repository (innovation-source) is to job postings, in terms of the skill-profile and concepts involved, then we can represent how this innovation-source has diffused over time (i.e. by the date of job posting). By relating this to the temporal development of the innovation-source, as represented by some other measure of the OSS repository’s history, we can the development-diffusion timeline of that particular innovation.

If true, this insight has tangible benefits for the practice forecasting skill demand in the AI sector, within which open-source software is a prolific source of innovation. Moreover, job postings & OSS data are high-frequency and accessible, meaning that more granular and fine-tuned predictions can be produced in effective real-time.

1.1 Thesis Argument

As a student of philosophy, I find that a strong start to any paper is to frame the content and argument clearly.

The function of this is to provide a clear view of what is either assumed to be true, what is *proven* to be true (as follows from the relevant literature), or what I expect to *demonstrate*. Thereafter, the conclusion necessarily follows, and it can be seen that the function of this paper is to prove the hypotheses as so labelled.

Argument 1.1: Thesis Argument

Assumptions

- A1** Open-source activity reflects experimentation.
- A2** Repo stars are, in large numbers, informative.
- A3** Only English language sources of text are considered.
- A4** The location of a job posting has a negligible effect.

Hypotheses

- H1** GitHub activity precedes job market demand.
- H2** AI-related skill mentions are represented semantically in similar ways between Job Postings and GitHub repository text, enabling cross-domain semantic comparison.

Theoretical Premises

- P1** There exist validated NLP methods for extracting technical skills from job postings with high precision and recall [2].
- P2** Contextual embeddings (e.g., Sentence-BERT) map job postings and repository descriptions into a shared feature space where cosine similarity reflects semantic similarity, as validated on semantic textual similarity benchmarks **reimers-gurevych-2019-sentence**.
- P3** Technical skills and repository descriptions are expressed in relatively standardized terminology, enabling reliable extraction and semantic comparison [3].
- P4** Stars indicate repository visibility and perceived popularity, while forks indicate community engagement and experimentation with the repository [4].
- P5** Sustained repository activity (commits, issues, pull requests) signals ongoing experimentation and adoption of new technologies **borges-2016-popularity**.
- P6** Early engagement with repositories may reflect emerging technology trends that are later reflected in labor market demand.
- P7** Job postings represent explicit employer demand for skills and thus serve as a proxy for labor market demand trends.
- P8** Temporal trends in job postings correlate with broader industry hiring trends **radlinski-2011-learning**, [2].
- P9** Skill mentions in job postings are described with relatively consistent terminology, enabling robust extraction using NLP techniques [3].
- P10** Cosine similarity between embeddings in the same feature space provides a quantitative measure of semantic similarity suitable for ranking items **mikolov-2013-distributed**, **reimers-gurevych-2019-sentence**.
- P11** Cosine similarity preserves approximate local neighborhoods: if two items are both highly similar to a shared reference embedding, they are likely to be semantically related to each other **ethayarajh-2019-contextual**.
- P12** Cosine similarity induces a partial ordering of job postings with respect to each repository: if $\cos(J_A, R) > \cos(J_B, R)$, then J_A is considered more semantically aligned with R than J_B , enabling

1.2 Aims & Objectives

1.2.1 Primary Aim

To develop and evaluate an AI-related skill demand forecasting methodology, integrating innovation signals (from Open-Source Software in particular) and using job ads as a proxy for skill demand.

1.2.2 Objectives

1. **Data Collection:** Collect & collate data for both channels in:
 - (a) Open-source Software (GitHub): *textual content & historic measure of development*
 - (b) Job Postings (e.g. LinkedIn, Indeed) - *content & date-posted*
2. **Representation:** Develop a high-quality representation method for integrating textual data from both channels, involving:
 - (a) **Build a Baseline:** A weak, keyword-based mechanism.
 - (b) **Advanced Representation:** An intelligent and robust method for representing text as embeddings.
 - (c) **Experimentation:** Fine tune and alter the components of embedding model, evaluating against the baseline.
3. **Time series:** Investigate Relationship of derived time series, looking at Granger causality values and Pearson correlations.
4. **Discuss:** Critically evaluate the derived results, With reflections on the benefits and limitations of the approach.

1.3 Roadmap

In the following chapter, I discuss related pieces of work and how their findings informed the paradigm for this project's implementation. Given the domain intersectional nature of this paper, works that are "related" include those on the topics of economic theory, natural language processing and AI itself.

Following this, the implementation of this project is detailed. Firstly, this concerns how data was acquired and preprocessed. Thereafter, the primary component of the pipeline is discussed - namely, the representation pipeline. Within this, I demonstrate both how the baseline model and the representation model are built,

discussing the involved components and how experimentation can be conducted to find optimal constructions. The subsequent section demonstrates the downstream task of these representation results - in the context of time series analysis, the findings of our representation pipeline are presented alongside the data of GitHub Stars. Having demonstrated the function of the entire pipeline, this next section therefore details the results and rationale of several sets of experiments.

What thereafter follows is a critical evaluation of the process, discussing the project's limitations and its successes, as well as the opportunities for future work. Finally, We discuss in the conclusion the outcomes and results of this paper, and their implications.

Chapter 2

Literature review

2.1 Introduction

Before embarking on implementing the project, extensive research was conducted to understand the various elements at play. The following depicts a list of questions that were devised in the research stage, and are answered thereafter. It is with this information that we can build a picture of what is supported (and what is overlooked) by existing research. This serves as a foundation for my work, giving precedent to the study of the thesis at hand.

As previously mentioned, the thesis argument in the Introduction Chapter grounds its premises on empirical grounds, according to the research thusly discussed. I shall make reference as appropriate to the premises in question.

1. AI Revolution & Labour shifts

- (a) What's happening with AI?
- (b) How does this compare to previous technological shifts?
- (c) How does this affect the workforce?

2. Labour Demand Analysis

- (a) How is labour demand forecasted?
- (b) Does AI pose a problem to these methods?
- (c) Are Job Postings useful here?

3. Innovation & Open-Source Software

- (a) What is Open Source software?
- (b) Is OSS a valuable innovation signal? Why?

- (c) How can we measure the adoption rate of OSS?

4. Methodological Techniques

- (a) How can we effectively represent semantic content quantitatively?
- (b) What are the considerations for comparing semantic representations of different domains?
- (c) To what extent can we infer relationships between two time-series?

2.2 Economic Background

2.2.1 Labour (Employment) Demand & Dynamics

Labor demand is defined as the number of people that employers are willing to employ at a given wage during a specific time period [1].

As a result of various economic phenomena, Labor demand is dynamic and frequently changes: Firm-level performance, competition and policies in minimum-wage or taxation are among the many factors that can influence a firm's capacity to employ. At the microeconomic (firm/individual) scale, much of the negative effects of labour demand equilibria changes Are expected in a free-market economy.

There are, however, systematic shocks that would affect the labor demand on a macroeconomic (economy-wide) sense. These can be deeply damaging to the proper functioning of that economy. Examples of such systematic shocks include war, pandemics or financial crises. Broadly, these shocks will have effects that extend to beyond labour demand shifts, since the economy is composed of an unfathomably complex network of cause-effect relationships.

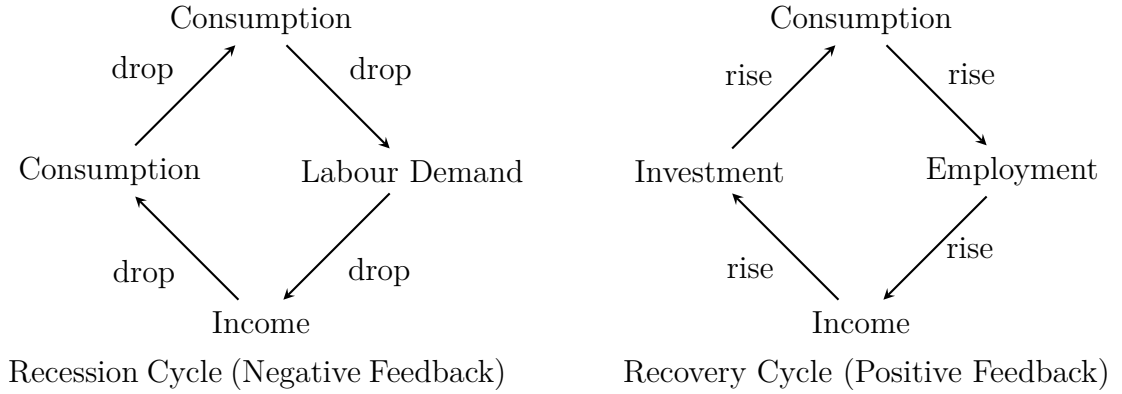
Such a consequence of this interconnectedness is the existence of feedback loops within the economy, as the standard Keynesian representation of recessions depicts [5] (see Figure 2.1):

2.2.2 Impact of Technological Innovation

Another driver of shifts in Labor demand is technology. Classical economic theory and some research suggests that technological development & adoption can be seen as a labor augmentation device, contributing positively to long term growth and employment [5], [6]. An opposing view, most famously represented by the example of the Luddites [7], depicts technological advancement as detrimental to employment prospects.

The authors of [8] assess and classify the meta-theories of technological innovation from classical texts and modern sources. Among their findings is that there

Figure 2.1: The inter-connected Economy



are a wide variety of views on the dynamics of technological innovation, each exhibiting different causal understandings as depicted by the authors' meta-theory classifications.

2.2.3 Role of Artificial Intelligence

Artificial intelligence is perhaps in A separate category of technological innovation for a few reasons. As is its nature, it is a different type of technology as it is inherently regarded as *intelligence*. This can be seen as both complementary and substitutional to human intelligence or labour [1].

This gives rise to emergent phenomena as it relates to technological innovation and labour:

1. **AI *as* innovation:** AI as a technology in the classical efficiency-mutliplier sense.
2. **AI *drives* innovation:** The authors of [9] examine the effects that AI has *on other* technological innovation, finding a strong positive relationship, especially in low-tech sectors.
3. **AI *replaces* innovation:** Artifical General Intelligence (AGI) is the abstract concept of a technology with an intelligence surpassing that of all humanity [1].

It is the 3rd point here that is of deep importance, because it represents such a profound shift a civilizational sense. Aside from the controversy and disagreement that surrounds the topic of AGI [1], it is clear that Labor demand will be profoundly impacted by the continual development of AI [1].

2.3 Labour Demand Forecasting

2.3.1 Overview

The practice of forecasting the demand (and *supply*) for labour has a long history [1]. Many critique its utility [10], but there is good reason to believe that the limitations faced by current approaches are not insurmountable. What follows is the findings of research into existing approaches, with an emphasis on their limitations.

2.3.2 Typological Approaches to Forecasting

2.3.2.1 Manpower Requirements Approach [10]

The authors of [10] discuss the *Manpower Requirements Approach (MRA)* for forecasting labour demand, within the context of Aboriginal Canadian communities. For context, the MRA approach is still central to the US’s Bureau of Labour Statistics (BLS) occupational-forecasting methodology, among other major institutions. In summary, the approach utilizes a “future macroeconomic reference scenario for GDP [...] to derive an estimate of total employment demand, which is later broken down by industry and by occupation.”

The authors modify the approach slightly, due to some concerns about feasibly producing “macroeconomic reference” scenarios as they concern the Aboriginal population of Canada. As a result, alongside a separate model they define for labour supply, their methodology is considered to be a fairly comprehensive framework for understanding demand & supply.

An important contribution they make is to lament the importance of forecasting not only “Expansion” demand (an increase in demand in absolute terms) but also “*Replacement*” demand (an increase in demand resulting from gaps created by mortality, retirements, occupational mobility, etc.).

However, the methodology is deeply assumption-led and is consequently limited.

- **Static occupational coefficients:** They assume that occupational distribution by industry remains constant over time
- **Zero elasticity of substitution between labour types:** workers in different occupations aren’t easily substitutable.
- **Extrapolation of trends:** e.g., labour force participation or enrolment rates follow past patterns—even if future social or structural shifts may alter them significantly.

Furthermore, as the authors recognize, the framework is fraught with other issues including:

- Data Limitation problems for smaller institutions
- An assumed heterogeneity of individuals within occupations misses a granular understanding of the individuals' differences (e.g. their skill-sets)

To add to data limitation problems, the authors find that a best practice is to use “up-to-date data and regularly refresh forecasts (every 2–3 years, medium-term horizon of 5–10 years)”. This is insufficient with respect to today's economy, especially considering the fast-paced shift in the labour market as a result of AI.

This review of labour forecasting methods is not without its issues, but highlights some important considerations for future endeavours.

The major limitations of their approach are:

1. Lacking granularity beyond occupation-level
2. The assumptions are highly simplified
3. Data requirements are extensive
4. Time-wise granularity is poor

Moreover, occupation-level findings of such forecasts are increasingly irrelevant, for reasons detailed later.

2.3.2.2 Quantitative Econometric Approaches [11]

Though outdated, the review [11] is a comprehensive discussion of the various methods available for institutions to produce labor market forecasts. Namely, they discuss with great detail methodological approaches following from *Econometrics*: the field of describing economic systems mathematically.

The predominantly discussed approach to demand-side forecasting is through econometric macroeconomic models: linking macro indicators (like GNP or industry-level output) to employment and occupational demand.

They observe that forecasting methods are not without limitations, among which are:

- Data problems
- Limits to understanding
- Past behaviour is not indicative

- Technical (Statistical) difficulties
- Inherently unpredictable events

Furthermore, they discuss and address explicitly that though approaches like MRA utilize an occupation-level granularity for forecasts, the lost nuance is detrimental to their defensibility. They recognize that policy-makers and governments will find two/three dimensional forecasts (relating to industry/occupation/qualification dimensions) increasingly irrelevant and incapable of producing informative forecasts in the face of those utilizing multidimensional data (“functions, tasks, abilities, personal attributes, characteristics and generic skills”).

Data Limitations and detailed understanding are the primary concerns that the authors raise as to the viability of econometric forecasts.

2.3.2.3 Other approaches: Survey-based, Input-Output

2.3.2.4 UK Context

Robert A. Wilson, the author of [11], contributes also to many specific forecasting efforts especially considering the UK government’s Working Futures programme (WFP). This is the primary tool that the UK government uses projecting future employment demand, and is very comprehensive.

The Working Futures Report [12], produced by the Institute for Employment Research (IER) of Warwick University [13] in partnership with other institutions (including the UK Commission for Employment and Skills (UKCES) and the Department for Education), is a “comprehensive, systematic, consistent and transparent set of projections” of the likely future’s labour market. Their projections are centrally grounded by projections of the E3ME Cambridge Econometric Model [14], producing multi-sectoral & multi-regional projections for 75 industries. Their rationale suggests that “sectors / industries [are] the key driver of changes in employment by occupation” [12]. Producing 10 year projections at intervals of 3 years, the report also makes use of with modules for replacement demand, occupations, and qualifications, with data from the Office for National Statistics (ONS) [15]. Framed not as exact predictions, the purpose of this forecast is to inform policy given outlooks or views of potential futures.

An immediately noticeable limitation of this approach is the periodicity of updates. For example the 2017 – 2027 forecast is produced in 2020. With a base year of 2017, the forecast can only be produced after a lag-time for the purpose of acquiring the relevant data, such as national figures, which are released at a lag of 1 – 2 years. Thus, the next report, grounded in 2020, would be expected to

release in 2023 - this most recent report is structured somewhat differently, with a longer time-horizon (2020 – 2035) [16].

To some extent, the report is immune to a critique of its long-term accuracy, given its framing as a “future likelihood” tool. Nevertheless, *improved* accuracy is always tangibly beneficial. Those critiques (as the report itself details) discuss those limitations:

- The forecast is heavily assumption-led
- The forecast results are not designed for precise local-area forecasts
- The forecast may lack the nuance of *intra*-occupational change

Further, a critique produced by [17] finds that many forecasts share the same limitations:

- Difficulty in linking from macroeconomic trends or employment forecasts to skills
- Few attempt to understand how skills change *within* occupations.

2.3.3 Alternative approaches

There are a number of solutions that are being investigated that aim to resolve some of the aforementioned limitations of labor market forecasting.

In the past, “naturally occurring data” (data not produced for analytical purposes) such as online job postings were difficult to assess systematically because of their lack of structure [18]. Furthermore, it has not historically been the case that vacancies are always advertised online, thereby adversely affecting their strength in representing labor market demand. However, these concerns are both increasingly irrelevant, due to advancements in Machine Learning techniques and the almost ubiquitous practice of advertising vacancies online.

These offer a potentially powerful data-source for understanding how the labor market is changing, and are under thorough investigation in many situations. One example is how the author of [19] uses Job Posting text to predict salaries, demonstrating the informative and predictive power of the information expressed by the natural text source.

Among those most pertinent to this paper are those that extract and portray the evolution of skill-requirements.

The Bank of England (BoE) employ an unsupervised Machine Learning (ML) techniques with job-posting data to identify disaggregation/segmentation methods for forecasts that are not at a sector/occupation level as in the WF Report,

but are instead at a skills-level [20]. They recognize the power of this Machine Learning based approach to devise data-driven taxonomies for skills that are otherwise unobserved by official classifications, which “cut across wage, sector and occupation”.

The European Centre for the Development of Vocational Training (CEDEFOP) have an ongoing project in their *Skills Online Vacancy Analysis Tool for Europe* (Skills OVATE), which uses vast data pipelines from 27 Nations (EU member states, EEA countries and the UK) to produce *quarterly* updates on a skill-demand forecast [21]. Another example of how skills can be investigated using job postings is presented by the OECD. They analyse Job-posting to produce an outlook on the demand for AI-related skills across sectors [22].

Other research suggests the power that Job Postings have to produce meaningful insights into how skills within occupations are changing [23]–[25].

2.3.4 Key Insights

The approaches available for a given economic actor to produce forecasts of labor market demand are varied. Whilst mostly different in their these methods, there are shared problems.

In general, for comprehensive models, high quality & voluminous data are needed at regular intervals, from a panoply of sources. This is deeply problematic not only for those private individuals with few resources available to them, but also to firms and even policy-makers, since acquiring such data introduces a necessarily longer lag-time for new forecasts. Furthermore, macroeconomic models experience difficulty in projecting intra-occupational change - this is fatal for properly understanding a labor market which is driven fundamentally by the more granular unit of individual skills, from which occupation-level requirements are composed.

As such, a variety of solutions are being investigated that take a more data-driven and bottom-up approach for understanding this more functional disaggregation of labor-demand - i.e. in terms of *skills*. Integrating the information observed in naturally-occurring data such as job postings to aid forecasts is ongoing, but they nevertheless show promise due to the informative, high-frequency and high-volume nature of job posting data, and are therefore worthy of consideration.

2.4 Open Source Software (OSS)

2.4.1 Open-source concept

Open-sourcing is a powerful concept within technological information. Before the internet, published information was centrally controlled, by the publishers or curators of such information. With the conception of internet tech, widespread collaboration on such information sources became viable, leading to openly available and extensively developed sources of information such as Wikipedia - this version of an encyclopedia is different to its paper-printed ancestors because *anyone* has the ability to propose and implement changes (under administrative audit). As a result, Wikipedia is the go-to encyclopedic source of information, known for its vast and detailed coverage of topics, as well as its reliability in presenting up-to-date and trustworthy information.

2.4.2 Open-source for Software

Open-source software (OSS) is the release of software under a license that permits (under varying conditions) the software's modification and redistribution [26]. On the whole, OSS is designed to promote collaboration and innovation within software, and is a paradigm under which many of the most powerful software tools are developed. Analogous to what Wikipedia is for general information, the most well known "host" of OSS is GitHub - this is a platform which provides extensive access to, and methods to collaborate with (or contribute to), OSS projects. There are other such platforms, but GitHub is by far the most popular. Central to what makes GitHub successful is its powerful version-control system (Git) - functionally, this is the mechanism by which multiple people can work on a document simultaneously without conflicting changes, as well as providing routes to revert changes and control permissions.

For the remainder of this paper, I shall use interchangeably the notions of OSS and GitHub:

Stipulation

The terms **GitHub (GH)** and **Open-Source Software (OSS) Platform** (and equivalently, **GH repositories** and **OSS products/projects**) are used interchangeably.

Though GitHub is not the *only* source-code host for OSS, it is by far the most popular [27]. Furthermore, given that the scope of this discussion is not that of GitHub (the platform) and instead concerns its hosted projects (i.e. Repositories), this stipulation is relevant because I intend to use GH Repositories as

proxies for OSS projects in general (The defensibility of this proxy assumption is examined under Section 9.1.1). This is an important distinction, and promotes pragmatic/effective discussion without unnecessary jargon.

2.4.3 OSS Importance

Many view that open-source is a practice that underpins the success and advancement of software technology. As the authors of [28] report, OSS should be treated as a public good. They further lament the unseen cost to developers within OSS, as well as specifying the necessity of upkeep for such ecosystems.

Indeed, given the expectations transformative power of AI, it is regarded open-sourcing relevant software is more important than ever [29], [30]. It is seen as vital that powerful software systems such as AI ought be democratized to prevent exploitation arising from private ownership, allowing the benefit to distribute more evenly among the population: “We believe the open source approach is the only way to achieve the full potential of AI, making it safer, accessible and democratized” [29].

2.4.4 OSS as Innovation

Open source software (OSS) is increasingly a rich ecosystem for The development of new software technologies. Often it is regarded as a hub for experimentation and innovation, and is a unique depiction of innovation outside the incentive structure observed by private economic agents.

Weak ties (low-intensity, infrequent interactions) are a long-theorized and recognized driver of innovation, since in a broad sense, they expose individuals to ideas from unrelated economic & social circles [8]. The authors of [31] investigate the impact of weak-ties, in the context of software development circles, on software innovation. Their findings suggest that a developer/contributor’s future project innovativeness is strongly correlated with the breadth and diversity of projects they engage with. This is highly suggestive of the power that the OSS ecosystem has to induce a positive externality-like effect for software innovation.

Similarly, the authors [32] examine “the imports of libraries in 12 different programming language ecosystems within millions of Stack Overflow posts over a 15 year period”. They aim to present learnings of OSS innovation to the literature on innovation dynamics, and find that recombination is a key innovation process in software (resembling concepts such as combinatorial innovation [33]). This is novel because they demonstrate the efficacy of treating the OSS ecosystem as similar to patenting and scientific research in terms of the innovation dynamics they express. The authors of [34] observe similar dynamical traits, doing so with

a novel approach of examining the dependency growth and complexity of GitHub repositories.

2.4.5 Quantifying OSS innovation

A key consideration for understanding the impact or proliferation of an OSS project, thereby in attempt to measure its innovative qualities, is defining how this is *measured*.

The authors of [35], [36] produce research that investigate the popularity growth of GitHub repositories. Their first effort “[identifies] four main patterns of popularity growth, which are derived after clustering the time series representing the number of stars of 2,279 popular GitHub repositories” [35]. Growth patterns identified are Slow, Moderate, Fast, and Viral. Expanding in later work [36], they:

1. expand their analysis to 5000 repositories,
2. survey 791 developers to identify the motivation / reasons behind starring,
3. investigate of factors influencing star-growth (e.g. marketing),
4. Develop a Machine Learning method to identify specific factors that predict a GH repository’s popularity growth.

2.5 NLP for Skill Extraction in Job Postings

2.5.1 Skill and Competence Taxonomies

To represent the “skill profile” of a Job Posting (JP) or a GitHub Repository (GHR), we need a **taxonomy** to define the concepts of a “skill”. Official classification bodies such as ESCO and O*NET exist for this broad purpose. They define hierarchical classifications to depict skills requirements for occupations.

The term *skill* itself is, however, only narrowly defined by the *European Skills, Competences, Qualifications and Occupations* (ESCO) classification [37] as “the ability to apply knowledge and use know-how to complete tasks and solve problems.” ESCO further distinguishes *knowledge* as “the outcome of the assimilation of information through learning,” encompassing facts, principles, theories, and practices related to a field of work or study, and defines *competences* as “the proven ability to use knowledge, skills, and personal, social, and/or methodological abilities in work or study situations and in professional and personal development.” Thus, competences capture the broader capacity to apply both knowledge and skills within specific contexts.

These classifications are problematic for the purposes of this project, because in practice, ESCO does not make a clear distinction between skills and competences within its “skills pillar”. ESCO also defines many items that are ability-wise relevant to AI (e.g. specific programming languages) as knowledge rather than skills. Additionally, ESCO classifications are manually defined, and may therefore be missing the nuance of recent & emergent taxonomies in the domain of AI. The strict and somewhat outdated ESCO classifications are therefore problematic for framing what the “skills” in this domain are.

Therefore, for the remainder of this paper, I shall stipulate the following:

Stipulation

The term “skill” is in this paper used broadly to encompass all three ESCO categories insofar as they relate specifically to AI technology.

This serves to simplify the discussion of how GitHub and Job Postings relate within the domain of AI technology, and should therefore not be confused with how ESCO defines a skill.

2.5.2 Background NLP

The simplest Natural Language Processing (NLP) technique is keyword matching: querying a body of text to determine the presence of a defined keyword. Though straightforward, This method is too narrow to capture the complexity of *natural* language, which encompasses alternative terms of phrase, acronyms and so on. To be useful, this method would involve curating a large dictionary of terminology, which is laborious.

An improvement would be to use fuzzy matching techniques, which allows for matching on typos and some small patterns alternative phrasing. However, this approach remains limited in its ability to capture *meaning*. It is not enough to simply define a match on a word-for-word basis, because a crucial element of what makes natural language meaningful is the context. For example the word “Python” alone means something very different when seen in the context of “Python, cobra and anaconda” as compared to “Python, SQL and Java”.

Building on some fundamental breakthroughs in the field of NLP, the seminal paper [38] depict a *Transformer architecture* that represents natural language with a contextual appreciation for each word (or token) in a sentence. With this, we can “embed” a body of text high dimensional feature space that meaningfully captures the text’s contextual and semantic properties. This technique therefore improves computational efficiency and reduces the manual oversight needed to process and depict what a piece of natural language is alluding to.

2.5.3 NLP for Skill-extraction

In [3], the authors develop a data driven methodology to extract and represent a job-skill taxonomy. Emphasizing that they do so independently of official boards for taxonomy definition (e.g. O*NET and ESCO), they Leverage NLP and network analysis methods for an algorithmic methodology to construct skill taxonomies, which remain adaptable and avoid expert elicitation. The significance of this is that it reinforces the feasibility of using NLP techniques, and specifically sentence embeddings, to extract skills from job postings.

Named Entity Recognition (NER) is a method of Natural Language Processing which identifies and extracts known entities (such as companies, people, or locations) from a body of text. The authors of [2] develop a specialized Named Entity Recognition (NER) Model that identifies and extracts the skills from within the body of text for a job posting. Their

2.6 Sentence Embeddings and Cross-Domain Semantic Comparison

2.6.1 Overview of Sentence Embeddings

A *Transformer* language model

What is a sentence transformer?

What are embeddings?

What are its advantages?

What are the limitations?

What's "similarity"?

As previously mentioned, the development in the field of Natural Language Processing was the Transformer architecture, which enriches the embeddings of tokens in a sentence with contextual meaning. **Somehow** sentence transformers distill a sequence of enriched token embeddings into a single embedding. **HOW?**

These sentence transformers therefore depict a sentence in terms of one high-dimensional vector of semantic features. Comparisons between vectors of different sentences are therefore comparisons of the values for each encoded feature of a sentence. Predicated on the fact that these feature spaces are sufficiently representative of semantic meaning, this equates to comparing the semantic meaning of two sentences.

From linear algebra, a method of compounding the distance between two vectors is the cosine distance - functionally, this measures the direction between two vectors, and is derived from their dot-product (adjusting for magnitudes). This

solution provides a meaningful comparison of the semantic similarity between two sentences as a number ranging from -1 to 1 .

2.6.2 Cross-Domain Applications

2.7 Synthesis and Research Gap

2.7.1 Integrated Understanding

2.7.2 Remaining Challenges

2.7.3 Justification of Current Approach

Chapter 3

Data Methods

3.1 Collection

In the case of this project, there are two facets of required data:

1. Open-Source Software Proliferation Data
2. Job Postings Data

3.1.1 Open-Source Software Data

Thankfully, the process of collecting data related to open-source software is fairly straightforward. As the name suggests, the OSS projects are intended to be *open* and accessible. This necessitates the usage of some online platform for hosting and maintaining the codebase, with strong version-control systems to safeguard a maintainable codebase. GitHub is functionally the default choice for hosting both open-source (and closed-source) codebases for a number of reasons, and is therefore the most relevant source for the required data of this project.

This project seeks data that more or less speaks to how *popular* a given GitHub repository is, and the prime candidate for this is the *GitHub Stars* mechanism. Users will "star" a repository in the same manner that internet users "bookmark" a website, behaving as a shortcut. This is a grossly simplified view of how GitHub repositories are interacted with, however - most GitHub repositories have no stars, and most users have "starred" very few repositories in return. The star is instead a useful proxy for measuring interest from a development point of view - if someone is contributing regularly to a repository, they will star it. In this sense, stars are a useful measure of how a repository is gaining interest from developers (and in a lesser sense, users), which in turn represents a growth in the utility the repository represents.

[39] is an open-source project that provides data on the historic star-counts of GitHub repositories, marketed as a tool to "provide insights into a repository's trendiness". Within the ML/AI framework that I delineated, the relevant repositories' star counts were found and downloaded thusly.

3.1.2 Job Posting Data

As is generally true of commercially valuable data, it can prove difficult to find open data that properly satisfies the desired criteria. Generally, it is a relatively simple task to build legally compliant scrapers that programatically access and store the data from job boards online. However, finding stores of historical data presents a major challenge, because only *active* job postings are present on job boards. This means that the scraper must be active throughout and is not useful for collecting data in retrospect.

An alternative solution is to therefore find and access a pre-prepared dataset. This, unfortunately, proved challenging in the sense that:

- Large, high-quality datasets are presented as products, and are thus expensive
- Small, low-quality datasets often lack essential features, such as "job description" and even "date"

The solution to these problems was to utilize *several* datasets that occupy different time-regions, made available on Kaggle. An overview of the characteristics for each dataset is included in the Appendix, as per Table ??.

At this rate,

Chapter 4

Representation

Chapter 5

Cross-Domain Similarity

Chapter 6

Time Series Methods

Chapter 7

Experiments

7.1 Evaluation Against Weak Model

Table 7.1: Embedding Evaluations (against Weak Baseline)

Run ID	Macro AP	MSE Weighted vs. Similarity	Micro AP (mean)	Micro NDCG10 (mean)	Micro P@1	Micro P@5	Micro P@10
v1_a	0.2320	0.1585	0.5583	0.5611	0.6184	0.5431	0.5215
v1_b	0.2208	0.1615	0.5639	0.5694	0.5623	0.5331	0.5475
v1_b_pca256	0.2494	0.1451	0.5400	0.5408	0.5666	0.5230	0.5181
v1_b_pca64	0.2474	0.1839	0.5400	0.5385	0.5593	0.5194	0.5163
v1_b_pca128	0.2541	0.1618	0.5399	0.5395	0.5574	0.5223	0.5177
v1_c	0.2127	0.1420	0.5429	0.5457	0.4968	0.5262	0.5325
v2_a	0.2194	0.2336	0.5526	0.5562	0.5249	0.5342	0.5406
v2_b	0.2035	0.1838	0.5505	0.5594	0.5307	0.5450	0.5452
v2_b_pca256	0.2342	0.1601	0.5417	0.5447	0.5352	0.5249	0.5245
v2_b_pca64	0.2320	0.1778	0.5435	0.5456	0.5417	0.5253	0.5241
v2_b_pca128	0.2329	0.1549	0.5420	0.5448	0.5398	0.5241	0.5238
v2_c	0.2175	0.2035	0.5532	0.5352	0.3788	0.5087	0.5368
v2_c_pca128	0.2345	0.1811	0.5329	0.5280	0.4923	0.5074	0.5172
v2_c_pca256	0.2358	0.1682	0.5323	0.5274	0.4912	0.5073	0.5167
v2_c_pca64	0.2343	0.1909	0.5340	0.5272	0.4906	0.5047	0.5164

Chapter 8

Discussion of Results

Chapter 9

Critical Evaluation

In this chapter (it may not be Chapter 4 for you, but probably Chapter 6 or 7 once all the core chapters have been added.)

The critical evaluation consists of a discussion, leading to conclusion. It is an essential part of a master's degree.

It shows that you can not only carry out a substantial piece of work, but that you can reflect on it, and think critically about how you might have done it better.

Examiners view the critical evaluation as very important.

Critical evaluation should contain

- Strengths and weaknesses of you project
- If you were unable to attain any deliverables, then why
- What are the future plans for your project if you are to continue

You will be presenting this during demonstration but here you need to discuss them in details.

9.1 Assumptions

9.1.1 Github repositories as proxies for OSS

9.1.2 Job Posting Source concatenation

Chapter 10

Conclusion

A brief summary of all that has gone before.

May include some directions for future work.

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Appendix A

A.1 Generative AI

This paper uses 3 Generative AI Tools:

1. **Cursor IDE** for expanding the weakly supervised model keyword dictionary, and implementing code efficiency improvements (using GPT-5) [40], [41]
2. **Elicit AI** to assist in literature exploration phase [42]
3. **ChatGPT** for advice, critique & education [41]

None of these tools were used to generate, or otherwise automate, the content of this paper (Except in the specific case of A.1.3.2) Underlying my use of generative AI was the imperative to avoid “Copy & Paste” strategies and instead use them as research, exploration and educational tools.

A.1.1 Elicit

For research, Elicit is a useful tool that performs a complex literature search based on detailed prompts. This was valuable for exploring research & informing the thesis more precisely.

I would, in future, make greater use of this tool for guiding the literature review stage.

Here’s are the *exact* prompts:

Prompt: Job Description Research

To what extent can we use the “Job description” section of Job Advertisements to map, forecast or understand the demand of certain skills or job requirements? A methodology that describes time-series representation of features in the Natural Language text, with NLP or other implementations.

Prompt: Innovation informed AI-Skill Forecast Research

Can innovation signal data (research papers, patents or open source software (e.g. GitHub)) improve the strength of skill demand forecasts derived from job postings (job boards like LinkedIn or Indeed), particularly for AI related skills? E.g. “prompt engineering” or “MLOps”

A.1.2 ChatGPT

For Code, ChatGPT was used for advice, debugging code or learning new programming frameworks. For writing, it was used as a critique assistant, to help explore ideas and critically assess the coherence of arguments or passages.

Here are some example prompts (adjusted, on a case-by-case basis):

Prompt: Conceptual Understanding

*Advise me on how to think about **{Concept/Tool/Framework/Method}** conceptually. I'd like to know the mental models or guiding intuitions that professionals use.*

Prompt: Conceptual Understanding

*Below is my summary of what I currently understand about **{Concept/-Tool/Framework/Method}**. Please critique it: point out what is accurate, what is incomplete, and where I've misunderstood. The goal is to refine my own understanding, not to receive a polished replacement text.*

Much of the time I use ChatGPT for workflow and tool advice, especially with regards to using Latex:

Prompt: Workflow advice

*I'm setting up a research workflow using **{Tool/Method}**. Can you suggest general strategies, potential pitfalls, and best practices? My goal is to design the workflow myself, but I'd like conceptual guidance.*

Prompt: L^AT_EXHelp

Advise me on strategies for managing a large dissertation in LaTeX. For example, what are best practices for splitting chapters into separate files, managing bibliographies, and handling figures/tables so that the project remains maintainable?

Prompt: Code Critique

*Below I include a piece of code. Rather than rewriting it for me, can you summarize your understanding of its purpose, critique its efficiency, point out any readability/-maintainability issues, and suggest what **principles** I should look at to improve it?*
{Code}

Prompt: Writing Critique

*I am writing a dissertation in **{Dissertation Context}**. I will paste a passage of my draft below. Please provide a detailed critique focusing on:*

- 1. clarity of expression and readability,*
- 2. academic style and tone,*
- 3. logical flow and argument structure, and*
- 4. any redundancy, ambiguity, or opportunities for conciseness.*

*Give only high-level feedback (structure, argument, coherence) - do not provide any specific rephrasing or line edits. Assume I am aiming for a professional, dissertation-quality standard. Here is the passage: **{Text Passage (No more than a paragraph at a time)}***

Each of the above prompts are used as guiding structures, and much of the time were adapted in scenario-specific follow-ups.

A.1.3 Cursor

A.1.3.1 Code Improvements

Cursor was used to advise on code improvements that were made strictly *after* a working solution was in place. The purpose of this was to improve both the computational efficiency and readability of the codebase. Further, it serves as an educational tool for improving my coding skills.

These adjustments were not made using the “Agent” feature of Cursor, but with “Ask”, such that it provides its suggestion for implementation at my discretion.

Here’s an example prompt for this use-case:

Prompt: Code Improvement Suggestions

*Within **{Context File}** in lines **{Context Lines}** is a piece of code. Currently it is functional. Rather than rewrite it for me, can you summarize your understanding of its purpose, critique its efficiency, point out any readability/maintainability issues, and suggest what **principles** I should look at to improve it?*

A.1.3.2 Weak-Model Keyword Generation

This is sole exception case where generated content was used: Instructing Cursor (using GPT-5) to construct the keyword dictionary for the weakly-supervised model. Its task here was to (In iterations of 50 items; 4 runs → 200 items):

- Expand a *Seed List* of keywords (Obtained by iteratively scraping GitHub topics)
- Match keywords to their commonly used acronyms
- Categorize the keywords

The grounding of this implementation in a *Seed List* was **instrumental** in making this effort defensible.

The purpose of this was to reduce manual & laborious data entry work - it should be noted that the only practical advantage of this implementation was to save time and effort. This entire process was audited and refined manually thereafter, to ensure output quality.

Prompt: Keyword List Curation

In {Context File} you'll find a list of keywords relating to Artificial Intelligence & related GitHub repository topics. Your task is to expand this keyword list, to a total of no more than 50. The resulting list should contain core concepts, tools, programming languages and domain-specific terms. If a keyword is commonly referred to by an acronym, these should be paired with its relevant verbose version. The acronym field should be left null for terms without commonly used acronyms. Be sure to preserve the casing for each keyword - e.g. "pytorch" should be "PyTorch". Make sure not to duplicate existing entries.

Prompt: Categorization

Within {Context File} are keywords that relate to artificial intelligence (including their acronyms). Create a new field of categories for each keyword into a set of no more than 25 different categories. These categories should be informative, and concise; if deemed necessary, one term can belong to multiple categories (no more than 3).

Data Sources

Third Party Code and Software Libraries

1. Whatever Software I end up using

A.1.4 ESCO Skill Classification

This publication uses the ESCO classification of the European Commission.

Table A.1: Overview of datasets used for Job Postings

Dataset Name	Date Range	Details	Source
1000 ML Jobs US	12/2023 - 04/2025	Mostly within March & April of 2025, all jobs are ML related	[43]
Linkedin Data Engineer Job Postings	17/12/2023 17/12/2023	- Extra "skills" column - This is appended to "description"	[44]
Data Science Job Postings & Skills (2024)	19/01/2024 21/01/2024	- Extra "skills" column - This is appended to "description"	[45]
LinkedIn Job Postings (2023 - 2024)	23/03/2024 20/04/2024	- —	[46]
Job Postings from Ireland (October 2022)	01/10/2022 31/10/2022	- Ireland only; Sample dataset from Techmap.io's commercial product	[47]
US Job Postings from 2023-05-05	05/05/2023 05/05/2023	- USA Only; Sample dataset from Techmap.io's commercial product	[48]

10.2 Word Counts

Category	Count
Words in text	6726
Words in headers	183
Words in captions	30
Tables/Figures	5
Inline math	22
Displayed math	0

Table 10.2: Document summary

File	Text	Headers	Captions	Floats	Inline	Displayed
					Math	Math
report	0	16	0	0	0	0
./envs/argument-environment	121	0	0	0	0	0
abstract	39	0	0	0	0	0
preface	171	0	0	0	0	0
introduction	763	9	0	0	0	0
literature	3820	98	0	0	15	0
core/1-data	450	9	0	0	0	0
core/2-representation	0	1	0	0	0	0
core/3-similarity	0	2	0	0	0	0
core/4-timeseries	0	3	0	0	0	0
core/5-experimentation	0	5	0	0	0	0
core/6-discussion	0	3	0	0	0	0
critical-evaluation	134	13	0	0	0	0
conclusion	16	1	0	0	0	0
appendices	1006	21	0	0	3	0
./figures/argument	327	0	0	0	4	0
./figures/feedback-loop	0	0	0	1	0	0

Table 10.3: Word count by file (`texcount`)